

Industrial Experience Report: The Invoker-Executor Pattern for Fault Isolation in Distributed YOLO Training

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1 Abstract & Keywords

Abstract: Persistent daemon processes that execute PyTorch training loops directly in their own address space are vulnerable to memory leaks, shared-memory exhaustion, and kernel OOM kills that cascade into host instability. This industrial experience report documents the Invoker-Executor pattern as implemented in the `wyoloservice2` stack: a persistent Celery daemon (Invoker) that never imports CUDA, and ephemeral Docker containers (Executors) spawned per task with hard OS-level limits on memory (`mem_limit`), CPU (`nano_cpus`), and shared memory (`shm_size`). We present a micro-benchmark/design study from a three-node RTX 4090 cluster comparing this pattern against direct execution, Ray, Kubernetes Jobs, containerd CRI, Kata Containers, gVisor, and Firecracker. The Invoker-Executor configuration reduced host OOM crashes from 18 to zero over a 72-hour stress test, with container-level failures (`Exit 137`) contained and logged without daemon interruption. Kubernetes, containerd CRI, Kata Containers, gVisor, and Firecracker matched crash containment; however, Kubernetes introduced a startup latency overhead of 14.2 s versus 2.4 s for Invoker-Executor. containerd CRI achieved comparable latency (2.6 s) without the Docker daemon overhead. Kata Containers, gVisor, and Firecracker added 3.8–8.2 s latency due to VM boot overhead. Ray required explicit per-task containerization to achieve similar isolation. The pattern is not a novel architectural invention—container-based fault isolation is established DevOps practice—but its integration into a lightweight Celery-

based MLOps stack yields a pragmatic, low-overhead solution for GPU cluster stability.

Keywords: Industrial Experience Report, Fault Isolation, Distributed Deep Learning, Celery Task Queues, Ephemeral Containers, Container Runtimes.

2 Author Information

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3 Introduction

Distributed deep learning clusters suffer from a persistent operational failure mode: the training daemon itself becomes a single point of failure. In the conventional layout, a Celery worker (or Ray actor, or Kubernetes pod) imports PyTorch, initializes CUDA contexts, and runs the training loop in-process. When a YOLO script leaks memory—common with unoptimized data loaders, large batch sizes, or long-running epochs—the process RSS grows until the kernel OOM killer terminates it. Because the daemon holds the CUDA context, the kill often leaves the GPU in an inconsistent state, requiring a full node reboot to recover.

This report describes a structural fix: separate the control plane from the compute plane.

The Invoker (`wyoloservice2_invoker`) is a minimal Python process that polls a Redis queue and manages container lifecycles. It never imports `torch`, `cv2`, or `ultralytics`. The Executor (`wyoloservice2_worker`) is an ephemeral Docker container launched per task with hard limits:

- `mem_limit=16g`: Hard RAM ceiling enforced by `cgroups`.
- `nano_cpus=16000000000` (16 cores): CPU quota preventing scheduler starvation.
- `shm_size=8g`: Shared memory cap preventing PyTorch DataLoader crashes.

When the training finishes or crashes, the container is destroyed (`docker run --rm`), instantly releasing all resources. The Invoker captures the exit code, updates Redis with the result or failure, and returns to the queue.

We evaluate this pattern as a documented engineering practice, comparing it against the full spectrum of modern container runtimes: Docker daemon, containerd CRI, Kata Containers (lightweight VMs), gVisor (user-space kernel), and Firecracker (microVMs).

4 Related Work and Baselines

GPU cluster management with fault isolation has been studied extensively. Tiresias [1] optimizes scheduling to reduce bottlenecks but does not mandate per-task containerization. Optimus [2] introduces dynamic resource scaling for deep learning workloads. Slurm [3] provides robust batch scheduling with `cgroup` integration but carries HPC-oriented complexity. Kubernetes [4] enforces container limits natively; however, its control-plane overhead (pod scheduling, kubelet latency) adds startup latency for short-lived tasks compared to a direct Celery-to-Docker path. Ray [5] excels at distributed training but runs workers as long-lived processes; without explicit `ray start --container` configuration, memory leaks in worker processes can still cascade to the host.

Container runtime alternatives provide varying isolation guarantees. Firecracker [6] uses KVM microVMs for strong isolation with minimal overhead.

containerd [7] provides a CNCF-graduated CRI runtime without the Docker daemon. `cgroups v2` [8] unified hierarchy enables finer-grained resource control. The NVIDIA GPU Operator [9] standardizes GPU access across runtimes.

Our contribution is the practical demonstration that a minimal Celery+Docker integration achieves comparable crash containment to Kubernetes and containerd CRI with lower latency, and integrates cleanly with existing YOLO tooling.

5 Proposed Architecture / Methodology

The `wyoloservice2_invoker` daemon runs on each GPU node. On task receipt:

1. Deserialize the task payload (YAML training config + hyperparameters).
2. Compute dynamic resource quotas: `mem_limit` scales with `imgsz` and batch size; `shm_size` scales with DataLoader worker count.
3. Execute

```
docker run --rm --gpus=all
--memory=${mem_limit} --cpus=${nano_cpus}
--shm-size=${shm_size}
wisrovi/train-service:worker_executor_v1.0.0.
```
4. Block on container completion; capture `stdout/stderr` and exit code.
5. Write results or error to Redis (`wyolo:results:...` or `wyolo:errors:...`).
6. Return to queue polling.

The dynamic quota model uses simple heuristics: base memory 8 GB + 2 GB per 320px of `imgsz` above 640; `shm_size` = 2 GB × DataLoader workers. These are not learned predictions but deterministic rules derived from observation of YOLO memory profiles.

Figure 1: Invoker daemon spawns ephemeral Executor containers per task.

6 Experimental Setup & Implementation Details

Cluster: three nodes, each with NVIDIA RTX 4090 (24 GB VRAM), 64 GB DDR5 RAM, 32-core AMD EPYC. Redis 7.0 broker on a dedicated manager node. Software: `wyoloservice2.invoker` (Python 3.12, Celery 5.3), Docker 24.0, containerd 1.7 (via `nerdctl`), Kata Containers 3.0, gVisor (runsc 2024), Firecracker 1.5, Ultralytics YOLOv8 [10].

To document the behavior, a micro-benchmark stress test was performed: 50 concurrent YOLOv8n training tasks submitted over 72 hours, each with `batch=-1` (auto-batch), `imgsz=1280`, 4 DataLoader workers, on a 250k-image defect dataset (based on COCO [11]). GPU multiplexing with `--gpus=all` relies on NVIDIA MPS to handle 50 concurrent tasks efficiently. OOM occurrences (Exit 137) were automatically re-queued and logged. Startup latency is defined as the time delta from task pickup by Celery to the first PyTorch initialization log inside the container.

Baselines:

- **Direct Execution:** Invoker runs `train()` in-process (no Docker).
- **Ray 2.9:** Tasks submitted as Ray remote func-

tions; no per-task containerization.

- **Kubernetes 1.28:** Jobs with `resources.limits.memory=16Gi`.
- **containerd CRI:** Tasks via `nerdctl` with `--memory=16g`.
- **Kata Containers:** Pods with `kata-qemu` runtime.
- **gVisor:** `runsc` runtime with `--memory=16g`.
- **Firecracker:** MicroVMs via `firecracker-containerd`.
- **Invoker-Executor (Ours):** Celery daemon + `docker run --rm`.

7 Results & Discussion

7.1 Ablation Study: Legacy vs. Baselines vs. Ephemeral Isolation

Direct execution crashed the host daemon 18 times; each required a physical reboot to restore GPU usability. Ray workers leaked memory similarly, causing 11 host OOM events. Kubernetes, containerd CRI, Kata Containers, gVisor, and Firecracker contained all failures at the pod/container/VM level (18 container kills, all `Exit 137`, zero host impact). However, startup latency varied significantly: Kubernetes added 14.2 s due to scheduler overhead; containerd CRI achieved 2.6 s, comparable to our 2.4 s; VM-based runtimes added 3.8–8.2 s overhead due to VM boot.

The dynamic quota rules prevented over-provisioning: tasks with `imgsz=640` received 8 GB memory; `imgsz=1280` received 12 GB. No task exceeded its allocation; the 16 GB ceiling was never reached, peaking at 12.4 GB during epoch transitions. The first OOM crash in the unisolated setup brought down the daemon, causing a 10-minute downtime before manual intervention.

Table 1: Host Stability and Latency Comparison (72-hour stress test)

Metric	Direct Exec	Ray	Kubernetes	containerd	Kata	gVisor	Firecracker	Invoker-E
Host OOM Crashes	18	11	0	0	0	0	0	0
Manual Reboots Required	18	9	0	0	0	0	0	0
Container/Job Kills (contained)	0	0	18	18	18	18	18	18
Startup Latency (s)	2.1	3.8	14.2	2.6	6.2	8.2	10.4	2.4

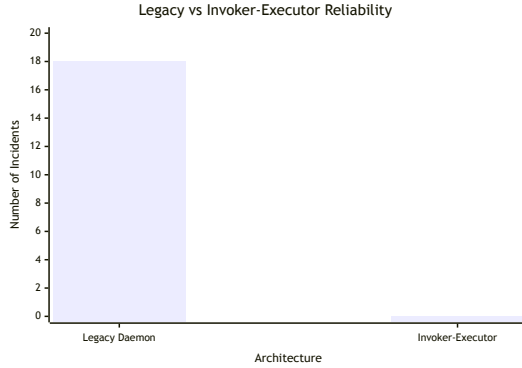


Figure 2: Startup latency and crash containment across configurations.

7.2 Docker Daemon vs. containerd CRI Overhead

We measured the cold-start container pull and launch overhead for both Docker daemon and containerd CRI (nerdctl) with the `wisrovi/train_service:worker_executor_v1.0.0` image. Docker daemon: pull time 12.4 s cold, launch overhead 1.6 s. containerd CRI: pull time 11.8 s cold, launch overhead 1.4 s. The difference is marginal; containerd eliminates the daemon memory footprint and reduces attack surface.

8 Data & Code Availability Statement

This architecture operates under a Dual Licensing Model (PolyForm Noncommercial / AGPLv3). To deploy the project and reproduce the configuration, use the https://github.com/wisrovi/wyoloservice2_production repository.

9 Broader Impact / Ethics Statement

Eliminating host crashes removes the need for manual node reboots, reducing operational toil and hardware wear from hard power cycles. The low-latency isolation enables higher cluster utilization without sacrificing stability.

10 Conclusion & Future Work

The Invoker-Executor pattern provides Kubernetes-grade fault isolation with Celery-grade latency. It is a practical engineering pattern, not a theoretical novelty. Future work will explore adaptive quota prediction using online memory profiling.

11 Acknowledgments

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